Amrish Yadav

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Summary

M.Tech graduate in Computer Science and Engineering (Data Science specialization) from IIIT Lucknow with proven expertise in SQL, Python, Power BI, and data engineering fundamentals. Experienced in transforming raw, multi-million record datasets into actionable business metrics through statistical analysis, automated KPI reporting, and dashboard development. Built enterprise-level shipment analytics frameworks at Amazon, optimizing forecast accuracy and stream reporting pipelines.

Technical Skills

Languages: Python, SQL, C++
Business Intelligence & Analytics: Power BI, Power Query, DAX, Advanced Excel, Key Performance Indicators (KPIs)
Data Analysis & Statistics: Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering
Databases & In-Memory Caching: MySQL, SQL Server, DynamoDB, Redis, Caching Strategies
Data Engineering & Streaming: Apache Kafka, AWS Athena, AWS S3, AWS Glue
Developer Tools & Frameworks: Git, GitHub, Jupyter Notebook, Docker, FastAPI, Streamlit, Postman

Professional Experience

Amazon - SDE Intern **July 2025 – December 2025**

- Developed a SQL-based analytics framework to analyze high-volume shipment data and support early shipment forecasting.
- Conducted detailed forecast bias analysis on historical shipment trends, achieving a 3.5% reduction in under-bias for 1-day early shipments and a 1.3% reduction for 2-day early shipments.
- Modernized legacy reporting services by migrating ingestion scripts to AWS Fargate, AWS S3, and AWS Athena, increasing reporting throughput and system scalability.
- Modernized a legacy reporting system by migrating reporting services to AWS Fargate, improving maintainability and scalability.
- Standardized core ETL logic and query execution templates across microservices, ensuring 100% metric consistency and improved data integrity across cross-functional dashboards.

Projects

Marketing Campaign Performance & ROI Analysis | *MySQL, SQL, Power BI, Power Query, DAX* **August 2026**

- Modeled three related campaign datasets covering 1,000 campaign records, 8 categories, and 6 regions; authored MySQL schema and SQL analyses for data quality, campaign performance, ROI ranking, and regional analysis.
- Recalculated spend-weighted net ROI as 65.56% on $ 42.54M revenue and $ 25.69M spend; flagged 235 of 1,000 records (23.5%) whose supplied ROI did not reconcile to revenue and spend.
- Built an interactive Power BI dashboard using Power Query and DAX to analyze campaign and regional performance, identifying Radio Ads as the higher-ROI channel (73.9% vs. 58.7% for Email Marketing) to inform marketing budget allocation.

End-to-End Credit Card Fraud Analytics | *Python, SQL, XGBoost, SHAP, FastAPI, Streamlit* **March 2026**

- Extracted and analyzed 283K+ financial transactions using SQL and Pandas, executing deep exploratory data analysis (EDA) to uncover key temporal and monetary fraud patterns.
- Formulated a machine learning detection pipeline yielding 0.8165 PR-AUC, 98.65% Precision, and 76.84% Recall, leveraging SHAP analysis for full model explainability.
- Deployed a real-time web interface using Streamlit and FastAPI with Redis caching for low-latency scoring of incoming transaction payloads.

Education

Indian Institute of Information Technology, Lucknow **July 2026**
M.Tech, Computer Science and Engineering ***CPI:8.21***

Lokmanya Tilak College of Engineering, Navi Mumbai **May 2023**
B.E, Computer Engineering ***CPI:8.25***

Certifications & Programs

Power BI Internship - Microsoft Elevate (AICTE) | *Data Cleaning, KPI, DAX, Data Modeling* **March 2026**
SQL Beginner to Advanced for Data Professionals | *SQL, Data Analysis, Query Optimization* **April 2025**
Python 101 for Data Science | *Python, NumPy, Pandas, Data Analysis* **February 2025**

Publications

Assessing the Performance of Imbalanced Classification by Class Improvement | *ICRMIR 2023* **March 2023**

- Published a conference paper on imbalanced classification, evaluating 5 resampling methods across 3 real-world datasets and achieving best F1/AUC scores of 0.99/0.99 for credit card fraud, 0.96/0.94 for customer churn, and 0.81/0.71 for heart disease.